Usage of Monte Carlo simulations for solving veridical-type paradoxes.

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Abstract

We used both numerical and analytical methods to solve veridical type paradoxes. A Monte Carlo simulation served as our numerical approach. The analytical techniques included conditional probability, Bayes' rule, and Bayes' rule with multiple conditions. We successfully solved all the veridical-type paradoxes and explained why such extensive discussion of the Monty Hall problem arose in the 90s. Additionally, we varied the Monty Hall problem by using different numbers of doors and prizes.

Keywords: Monte Carlo simulation, Bertrand's box paradox, Three prisoners dilemma, Monty Hall problem, Bayes' rule, Bayes' rule with multiple conditions, conditional probability, veridical type paradox

Classification: 60, 62, 65

Introduction

A paradox is a statement that contradicts itself yet might be true or false simultaneously. A veridical paradox produces an outcome that seems absurd but is proven to be true nonetheless. We will focus on veridical paradoxes using Monte Carlo simulations, but we will also provide analytical solutions where possible. The key part of our research is the Monty Hall problem, and we primarily use Monte Carlo simulations for most calculations, although we will mention the analytical solution as well.

We will describe and solve the following veridical type paradoxes: Bertrand's box paradox, the Three Prisoners' dilemma, and the Monty Hall problem. The key part of our research was the Monty Hall problem, and we will explain why so many PhD researchers insisted on the probability being 0.5 back in the 90s.

1.1 Monte Carlo simulation

Monte Carlo simulation is an advanced statistical tool. A simulation is an experiment, typically run on a computer, that involves using random numbers. A stream of random numbers is a sequence of statistically independent random variables usually uniformly distributed in the interval [0,1).

Simulations are used when it is difficult to rely solely on analytical methods to model real-world situations or solve the underlying mathematical problems. They depend on repeated random sampling to produce numerical results. We used random sampling to estimate probabilities.

1.2 Bertrand's box paradox

Bertrand's box paradox is a paradox of elementary probability theory. There are three boxes:

1) A box containing two gold coins
2) A box containing two silver coins
3) A box containing one gold coin and a silver coin

The question is: What is the probability of choosing a gold coin, given that the first coin is also gold? The player is selecting a box at random and does not swap boxes after the first toss. This is a simple example of conditional probability.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (1)$$

P(A) – probability of choosing a gold coin in the second toss
P(B) – probability of choosing a gold coin in the first toss

$$P(B) = P(gold|GG) + P(gold|SS) + P(gold|SG) \quad (2)$$

$$P(B) = \frac{1+0+\frac{1}{2}}{3} = \frac{1}{2} \quad (3)$$

$$P(A \cap B) = \frac{1}{3} \quad (4)$$

$$P(A|B) = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3} \quad (5)$$

The result appears to be $\frac{1}{2}$ using common sense, but it is $\frac{2}{3}$ in fact. The correct solution to the problem has been well known for a long time. Our research aimed to build a Monte Carlo simulation of the problem. Monte Carlo simulation is an advanced statistical tool. We coded the following code in R:

```
#Bertrand's paradox
set.seed(100)
samplesize<-1000000
a<-sample(0:2,samplesize,replace=T)
# 3 boxes: 0- gold, gold; 1 - silver, silver; 2 - gold, silver
b<-sample(0:1,samplesize,replace=T) # 2 balls in each box
data<-data.frame(a,b)
data2<-subset(data,(a==0) | (a==2 & b==0),select=a)
round(sum(a==0)/nrow(data2),4) # final probability
```

Monte Carlo simulation confirms the probability equals 0.6667, which is $\frac{2}{3}$.

1.3 Three prisoners problem

The Three Prisoners Problem is another veridical type paradox. Three prisoners A, B, and C, are in separate cells and sentenced to death. The governor has selected one of them at random to be pardoned. The warden knows which one is pardoned, but is not allowed to tell. Prisoner A begs the warden to reveal the identity of the lucky one. Prisoner A knows that the warden can’t tell the identity of the one to be pardoned, so he proposed that the warden follow the code: “If B is pardoned, give me C’s name. If C is pardoned, give me B's name. If I am pardoned, flip a coin to decide whether to name B or C." The warden tells A that it will be B. Prisoner A is pleased because he believes his probability of surviving has increased from $\frac{1}{3}$ to $\frac{1}{2}$, as it is now between him and C. This is what common sense tells. The question is: what are the true probabilities? Analytical solution :

- A, B, and C are corresponding prisoners
- a, b, c are events in which the warden mentions that the corresponding prisoners were pardoned
- P(A),P(B),P(C) are probabilities that governor pardoned corresponding prisoners

$$P(A) = P(B) = P(C) = \frac{1}{3}, P(b|A) = \frac{1}{2}, P(c|A) = \frac{1}{2} \tag{6}$$

$$P(b|C) = 1, P(c|B) = 1, P(c|C) = 0, P(b|B) = 0 \tag{7}$$

It is a more complicated problem. It can be solved using Bayes' rule.

$$P(A|b) = \frac{P(b|A)\times P(A)}{P(b|A)\times P(A)+P(b|B)\times P(B)+P(b|C)\times P(C)} \tag{8}$$

$$P(A|b) = \frac{\frac{1}{2}\times\frac{1}{3}}{\frac{1}{2}\times\frac{1}{3}+0\times\frac{1}{3}+1\times\frac{1}{3}} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3} \tag{9}$$

$$P(A|c) = \frac{P(c|A)\times P(A)}{P(c|A)\times P(A)+P(c|B)\times P(B)+P(c|C)\times P(C)} \tag{10}$$

$$P(A|c) = \frac{\frac{1}{2}\times\frac{1}{3}}{\frac{1}{2}\times\frac{1}{3}+1\times\frac{1}{3}+0\times\frac{1}{3}} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3} \tag{11}$$

$$P(C|b) = \frac{P(b|C)\times P(C)}{P(b|A)\times P(A)+P(b|B)\times P(B)+P(b|C)\times P(C)} \tag{12}$$

$$P(C|b) = \frac{1\times\frac{1}{3}}{\frac{1}{2}\times\frac{1}{3}+0\times\frac{1}{3}+1\times\frac{1}{3}} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3} \tag{13}$$

The conclusion of the Three Prisoners Problem is that the warden's information does not reveal anything about prisoner A's future. The probability of being pardoned stays at $\frac{1}{3}$. The probability of being pardoned for prisoner C is now $\frac{2}{3}$.

Monte Carlo simulation has been coded in R:

```
# Monte Carlo simulation
# Three prisoners problem
set.seed(100)
samplesize<-1000000
governor<-sample(0:2,samplesize,replace=T)
fm<-function(a,j) {
if (a[j]==0) {thisone<-sample(1:2,1,replace=T)}
if (a[j]==1) {thisone<-2}
if (a[j]==2) {thisone<-1}
thisone
}
warden<- sapply(1:samplesize, function(j) fm(governor,j))
r<-data.frame(governor,warden)
# Warden told B, given governor choose A.
sum(r$warden==1 & r$governor==0)/sum(r$warden==1)
# Warden told C, given governor choose A.
sum(r$warden==2 & r$governor==0)/sum(r$warden==2)
```

```
# Warden told B, given governor choose C.
sum(r$warden==1 & r$governor==2)/sum(r$warden==1)
# Warden told C, given governor choose B.
sum(r$warden==2 & r$governor==1)/sum(r$warden==2)
```

Monte Carlo simulation confirms probabilities are 0.332 and 0.667, which is the same result as the analytical solution provides.

1.4 Monty Hall problem

The Monty Hall problem was the key part of our research. We will try to explain the discussion of the issue from the 90s.

Suppose you are on a game show, and you are given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what is behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

Common sense tells us that whether you switch your choice or not, you still have a 50 percent chance of winning.

Analytical solution :

Events C_1, C_2, C_3 are indicating the car is behind door 1,2 or 3.

$$P(C_1) = P(C_2) = P(C_3) = \frac{1}{3}$$

Event X_1 is indicating the player initially choosing door 1.

As the position of the car is independent of the player's first choice $P(C_i|X_1) = \frac{1}{3}$.

H_3 is the host opening door 3.

The following probabilities are obvious :

$$P(H_3|C_1, X_1) = \frac{1}{2}, P(H_3|C_2, X_1) = 1, P(H_3|C_3, X_1) = 0 \tag{14}$$

The probability that the car is behind door No. 2, given the player initially choosing door 1 and the host opening door No. 3, is :

$$P(C_2|H_3, X_1) = \frac{P(H_3|C_2,X_1) \times P(C_2|X_1)}{P(H_3|X_1)} \tag{15}$$

$$P(C_2|H_3, X_1) = \frac{P(H_3|C_2,X_1) \times P(C_2|X_1)}{P(H_3|C_1,X_1) \times P(C_1|X_1) + P(H_3|C_2,X_1) \times P(C_2|X_1) + P(H_3|C_3,X_1) \times P(C_3|X_1)} \tag{16}$$

$$P(C_2|H_3, X_1) = \frac{1 \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + 1 \times \frac{1}{3} + 0 \times \frac{1}{3}} = \frac{2}{3} \tag{17}$$

The probability that the car is behind door No.1, given the player initially choosing door 1 and the host opening door No. 3 is :

$$P(C_1|H_3, X_1) = \frac{P(H_3|C_1,X_1) \times P(C_1|X_1)}{P(H_3|X_1)} \tag{18}$$

$$P(C_1|H_3, X_1) = \frac{P(H_3|C_1,X_1) \times P(C_1|X_1)}{P(H_3|C_1,X_1) \times P(C_1|X_1) + P(H_3|C_2,X_1) \times P(C_2|X_1) + P(H_3|C_3,X_1) \times P(C_3|X_1)} \tag{19}$$

$$P(C_1|H_3, X_1) = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + 1 \times \frac{1}{3} + 0 \times \frac{1}{3}} = \frac{1}{3} \tag{20}$$

The ‚Flip a coin' decision has the following probability :

$$P(flip\ a\ coin) = \frac{1}{2} P(C_1|H_3, X_1) + \frac{1}{2} P(C_2|H_3, X_1) \tag{21}$$

$$P(flip\ a\ coin) = \frac{1}{2}\frac{1}{3} + \frac{1}{2}\frac{2}{3} = \frac{1}{2} \tag{22}$$

The analytical solution shows that the player should switch her choice, since the chance to win is $\frac{2}{3}$. In case the player does not switch her choice, the chance to win is just $\frac{1}{3}$. We used Bayes' rule with multiple conditions. Since the Monty Hall problem is a key part of our research, we have focused on many different strategies, which were simulated in Monte Carlo simulations.
We explored the following strategies :

- Not switching choice
- Switching choice
- Flip a coin decision
- Tic–toc strategy
- Opposite tic–toc strategy

We coded simulations that vary the number of doors and also the number of prizes. If the contestant decides to switch doors and more than one door is available, she will choose another door randomly. If the contestant flips a coin, she does it just once. If there is more just one door available, she will select another door randomly, too. The following code simulates different door counts and also car counts options. Code has been written in R:

```
#Monty Hall problem
samplesize<-100000 # sample size
doors<-9 # doors count -1
door<-vector("numeric",length=doors*(doors-1)/2)
prize<-vector("numeric",length=doors*(doors-1)/2)
changedp<-vector("numeric",length=doors*(doors-1)/2)
nochangep<-vector("numeric",length=doors*(doors-1)/2)
ttp<-vector("numeric",length=doors*(doors-1)/2)
ottp<-vector("numeric",length=doors*(doors-1)/2)
flipp<-vector("numeric",length=doors*(doors-1)/2)
results <- data.frame(door, prize, changedp, ttp,flipp,
ottp,nochangep)
#tic-toc oppposite tic-toc function
ttott<-function(ttott1,win1,win2,trigger){
if (ttott1==0) {if (win2==0) {thisone<-0
} else {thisone<-1 }}
if (ttott1==1) {if (win1==0) {thisone<-1
} else {thisone<-0 }}
if (trigger=="ott") {if (thisone==0) {thisone<-1
} else {thisone<-0 }}
thisone}
for (i in 2:doors) {
for (j in 1:(i-1)) {
set.seed(100)
initial<-replicate(samplesize,list(sample(0:i,1,replace=F)))
priz<-replicate(samplesize,list(sample(0:i,j,replace=F)))
# initial guess & prizes behind doors
flip<-sample(0:1,samplesize, replace=T) # flip a coin
tt<-sample(0,samplesize,replace=T) # tic toc
ott<-sample(0,samplesize,replace=T) # opposite tic toc
choices<-replicate(samplesize,list(c(0:i)))
# WHICH GOAT WILL BE SHOWN
```

```
goats<-mapply(setdiff,choices,priz)
goats2<- lapply(1:ncol(goats), function(p) goats[,p])
# goats are opposite prizes
notinitial<-mapply(setdiff,choices,initial)
notinitial2<-lapply(1:ncol(notinitial),
function(p) notinitial[,p])
# group of not initial decisions
goats3<-mapply(intersect,goats2,notinitial2)
goats4<-lapply(goats3, function(p) c(p,p))
goat<-lapply(goats4, function(p) sample(p,1))
# to show just one goat from intersect not initial & goats
remove(goats,goats2,choices,notinitial,goats3,goats4)
chmind<-mapply(setdiff,notinitial2,goat)#to change mind
#to exclude goat which was shown from not initial group
if (is.null(ncol(chmind))==FALSE) {
chmind2<- lapply(1:ncol(chmind), function(p) chmind[,p])
} else {chmind2<-chmind }
chmind3<-lapply(chmind2, function(p) c(p,p))
newmind<-lapply(chmind3, function(p) sample(p,1))
# to choose 1 new decision from all the available
remove(notinitial2, goat,chmind,chmind2,chmind3)
win1<-mapply(intersect,newmind,priz)#win1 changed mind
win2<-mapply(intersect,initial,priz)#win2 not changed mind
win13<-as.numeric(lapply(win1,function(p) length(p)==0))
win23<-as.numeric(lapply(win2, function(p) length(p)==0))
# intersection - 1 no intersection,0 intersection
for(l in 1:(samplesize-1)) { # TIC-TOC, OPPOSITE TIC TOC
tt[l+1]<-ttott(tt[l],win13[l],win23[l],'tt')
ott[l+1]<-ttott(ott[l],win13[l],win23[l],'ott')}
#PROBABILITIES
win1p<-sum(win13==0)/samplesize #win1p changed mind
win2p<-sum(win23==0)/samplesize #win2p not changed mind
remove(newmind,priz,initial,win1,win2)
flipfr<-data.frame(win13,win23,flip,tt,ott)
flip1<-nrow(flipfr[flipfr$flip==1 & flipfr$win13==0,])
flip2<-nrow(flipfr[flipfr$flip==0 & flipfr$win23==0,])
# flip1 - changed mind,flip2 - initial decision
tt1<-nrow(flipfr[flipfr$tt==1 & flipfr$win13==0,])
tt2<-nrow(flipfr[flipfr$tt==0 & flipfr$win23==0,])
ott1<-nrow(flipfr[flipfr$ott==1 & flipfr$win13==0,])
ott2<-nrow(flipfr[flipfr$ott==0 & flipfr$win23==0,])
flipp<-(flip1+flip2)/samplesize
tictocp<-(tt1+tt2)/samplesize
opptictocp<-(ott1+ott2)/samplesize
results$door[i*(i-1)/2-i+1+j]<-i+1#RESULTS - WRITTING
results$prize[i*(i-1)/2-i+1+j]<-j
results$changedp[i*(i-1)/2-i+1+j]<-round(win1p*100,3)
results$ttp[i*(i-1)/2-i+1+j]<-round(tictocp*100,3)
results$flipp[i*(i-1)/2-i+1+j]<-round(flipp*100,3)
results$ottp[i*(i-1)/2-i+1+j]<-round(opptictocp*100,3)
results$nochangep[i*(i-1)/2-i+1+j]<-round(win2p*100,3)
```

```
remove(win13,win23,flipfr,flip,tt,ott) } }
nname<- paste(toString(samplesize),"r.txt",sep="")
write.table(results,nname,append=FALSE)
```

The output of the codes is shown in Table 1, and 2. Probabilities were calculated with a sample size of 1.5 million. Probability theory allows a simple derivation of Monte Carlo simulation error. If the contestant does not change her mind, the probability of winning is a very simple formula.

$$P(not\ switching\ decision) = \frac{prizes\ count}{doors\ count} \tag{23}$$

We can measure the Mean Absolute Error of Monte Carlo simulations probabilities for the not-switching decision and Bias.

$$MAE = \frac{1}{36}\sum_{i=1}^{36}\left|p_{MC_i} - p_{true_i}\right| = \frac{0.933}{36} = 0.026 \tag{24}$$

$$Bias = \frac{1}{36}\sum_{i=1}^{36}\left(p_{MC_i} - p_{true_i}\right) = -\frac{0.191}{36} = -0.005 \tag{25}$$

Table No.1 *Simulations results*

No	Doors	Prizes	Changedp	Nochangep	Flip	Ttp	Ottp
1	3	1	0.6664	0.3336	0.4999	0.5552	0.4448
2	4	1	0.3752	0.2501	0.3123	0.3183	0.3000
3	4	2	0.7504	0.4997	0.6253	0.6660	0.5999
4	5	1	0.2665	0.2003	0.2331	0.2352	0.2290
5	5	2	0.5329	0.4001	0.4666	0.4752	0.4573
6	5	3	0.8001	0.5999	0.7000	0.7336	0.6854
7	6	1	0.2087	0.1666	0.1876	0.1881	0.1852
8	6	2	0.4163	0.3330	0.3751	0.3777	0.3704
9	6	3	0.6246	0.5003	0.5623	0.5715	0.5559
10	6	4	0.8333	0.6665	0.7499	0.7774	0.7405
11	7	1	0.1716	0.1426	0.1568	0.1572	0.1558
12	7	2	0.3427	0.2854	0.3140	0.3154	0.3114
13	7	3	0.5141	0.4294	0.4716	0.4753	0.4683
14	7	4	0.6852	0.5713	0.6284	0.6371	0.6237
15	7	5	0.8570	0.7143	0.7854	0.8095	0.7792
16	8	1	0.1461	0.1252	0.1355	0.1356	0.1349
17	8	2	0.2916	0.2498	0.2710	0.2716	0.2691
18	8	3	0.4375	0.3754	0.4064	0.4081	0.4040
19	8	4	0.5832	0.5000	0.5415	0.5457	0.5386
20	8	5	0.7290	0.6253	0.6774	0.6855	0.6736

Table No. 2 *Simulations results*

No	Door	Prizes	Changedp	Nochangep	Flip	Ttp	Ottp
21	8	6	0.8750	0.7498	0.8121	0.8336	0.8072
22	9	1	0.1272	0.1112	0.1191	0.1190	0.1186
23	9	2	0.2540	0.2218	0.2379	0.2383	0.2369
24	9	3	0.3809	0.3337	0.3574	0.3580	0.3561
25	9	4	0.5079	0.4449	0.4766	0.4781	0.4740
26	9	5	0.6351	0.5554	0.5951	0.5993	0.5924
27	9	6	0.7613	0.6667	0.7136	0.7221	0.7111
28	9	7	0.8889	0.7784	0.8340	0.8518	0.8301
29	10	1	0.1130	0.1000	0.1064	0.1061	0.1059
30	10	2	0.2251	0.1998	0.2127	0.2128	0.2114
31	10	3	0.3377	0.3003	0.3191	0.3194	0.3175
32	10	4	0.4501	0.4002	0.4250	0.4264	0.4236
33	10	5	0.5626	0.5000	0.5314	0.5338	0.5293
34	10	6	0.6750	0.5998	0.6371	0.6413	0.6347
35	10	7	0.7872	0.7006	0.7439	0.7511	0.7414
36	10	8	0.8998	0.8000	0.8501	0.8667	0.8466

Mean Absolute Error is 0.026, and Bias is -0.005. These figures are also approximations of the Monte Carlo simulations‘ error and Bias for the entire tables 1 and 2.
Taking the true probabilities for ‘not switching' decision into account, we can also reestimate ‘flip a coin' decision probabilities. They are in the ‘flipp combined‘ column for that decision.

$$P(flip\ a\ coin) = \frac{1}{2}P(switching_{Monte\ Carlo}) + \frac{1}{2}P(no\ switch_{true}) \tag{26}$$

Mean Absolute Error and Bias for this modification of the ‘flip a coin' decision are:

$$MAE_{modified} = \frac{1}{36}\sum_{i=1}^{36}\left|p_{MC_i} - p_{true_i}\right| \tag{27}$$

$$p_{MC_i} = \frac{1}{2}P\left(switching_{Monte\ Carlo_i}\right) + \frac{1}{2}P\left(no\ switch_{true_i}\right) \tag{28}$$

$$p_{true_i} = \frac{1}{2}P\left(switching_{true_i}\right) + \frac{1}{2}P\left(no\ switch_{true_i}\right) \tag{29}$$

$$MAE_{modified} = \frac{1}{36}\sum_{i=1}^{36}\left|\frac{1}{2}p_{switching_{MC_i}} - \frac{1}{2}p_{switching_{true_i}}\right| \tag{30}$$

Since (24) and (25) are approximations of MAE and Bias for the whole table 1 and table 2:

$$MAE_{modified} = \frac{1}{2}MAE \tag{31}$$

$$Bias_{modified} = \frac{1}{2}Bias \tag{32}$$

Modified probabilities show table 3 and table 4.

Table 3 *Combined results*

No	Door	Prize	Changep	No changep Monte Carlo	No changep True	Flipp	Flipp combined
1	3	1	66.645	33.355	33.333	50.061	49.989
2	4	1	37.545	24.992	25	31.278	31.273
3	4	2	75	49.993	50	62.54	62.5
4	5	1	26.704	19.979	20	23.35	23.352
5	5	2	53.287	40.047	40	46.657	46.644
6	5	3	80.014	59.99	60	69.997	70.007
7	6	1	20.86	16.649	16.667	18.791	18.764
8	6	2	41.67	33.303	33.333	37.493	37.502
9	6	3	62.565	50.053	50	56.253	56.283
10	6	4	83.273	66.719	66.667	75.042	74.97
11	7	1	17.155	14.286	14.286	15.767	15.721
12	7	2	34.268	28.581	28.571	31.477	31.42
13	7	3	51.457	42.892	42.857	47.114	47.157
14	7	4	68.592	57.163	57.143	62.917	62.868
15	7	5	85.661	71.399	71.429	78.553	78.545
16	8	1	14.564	12.49	12.5	13.562	13.532
17	8	2	29.171	25.009	25	27.097	27.086
18	8	3	43.77	37.492	37.5	40.609	40.635
19	8	4	58.368	49.989	50	54.252	54.184
20	8	5	72.953	62.504	62.5	67.716	67.727
21	8	6	87.511	74.955	75	81.251	81.256

The *Changedp* column describes probabilities in case the player changed her mind.The *Nochangep* column describes probabilities in case the player did not change her mind. The *flipp* column describes probabilities in case the player made a 'flip a coin decision'. The *ttp* column describes probabilities in case the player is applying the 'tic-toc' strategy. The *ottp* column describes probabilities in case the player is applying the 'opposite tic-toc' strategy.

'Tic-toc' strategy is a kind of strategy when the player makes a decision based on the last known decision. If the previous decision is to change mind and was successful, she will also change her mind. If the last decision is to change her mind and was unsuccessful, she will not change her mind. 'Opposite tic-toc' strategy is a kind of strategy where the player also makes a decision according to the last known decision. If the last decision is to change her mind and was successful, she will not change her mind. If the last decision is to change her mind and was unsuccessful, she will change her mind. She will do the opposite because she expects the situation will change in the next turn.

Another interesting fact is that the probabilities increase as the price count rises. Tables No. 1 and No. 2 show that changing your mind is the best strategy a player can use. 'Flip a coin' decision probabilities always fall between 'change mind' probabilities and 'do not change mind' probabilities.

Table No. 4 *Combined results*

No	Door	Prize	Changep	No changep Monte Carlo	No changep True	Flipp	Flipp combined
22	9	1	12.709	11.063	11.111	11.919	11.91
23	9	2	25.378	22.286	22.222	23.85	23.8
24	9	3	38.05	33.302	33.333	35.62	35.692
25	9	4	50.793	44.429	44.444	47.603	47.619
26	9	5	63.514	55.543	55.556	59.574	59.535
27	9	6	76.224	66.624	66.667	71.459	71.446
28	9	7	88.856	77.803	77.778	83.35	83.317
29	10	1	11.263	9.974	10	10.663	10.632
30	10	2	22.496	20.021	20	21.273	21.248
31	10	3	33.796	29.957	30	31.856	31.898
32	10	4	44.983	40.009	40	42.497	42.492
33	10	5	56.281	49.971	50	53.151	53.141
34	10	6	67.475	59.945	60	63.734	63.738
35	10	7	78.761	69.99	70	74.413	74.381
36	10	8	90.008	79.949	80	84.967	85.004

'Flip a coin' decision for three doors and one prize is those 50 percent, which caused a lot of discussion about the Monty Hall problem. Those 50 percent are those 50 percent, what common sense tells that the probability should be. 'Tic-toc' decision is better than a 'flip a coin' decision. 'Opposite tic-toc' decision is worse than 'flip a coin' decision, but better than 'do not change mind' at all. If the player decides not to change her mind, it is the worst decision ever. 'Flip a coin' decision is the third best decision of five different strategies, and it reveals the old truth: "If you do not know what you should do, make a 'flip a coin' decision, and you will not make a bad decision". Research also showed that people should change their minds, because individuals who do not change their minds have the lowest probability of success. The second worst chance for success have individuals who are speculating too much – 'opposite tic-toc' decision. The best chance for success have individuals who change their minds.

Finally, we can plot the dependence between the Monte Carlo simulations' sample size and Mean Absolute Error as *MAE* = *f*(*sample size*), and also the dependence between the Monte Carlo simulations' sample size and Bias as *Bias* = *f*(*sample size*). Dependence shows Figure 1 and Figure 2.

Figure 1

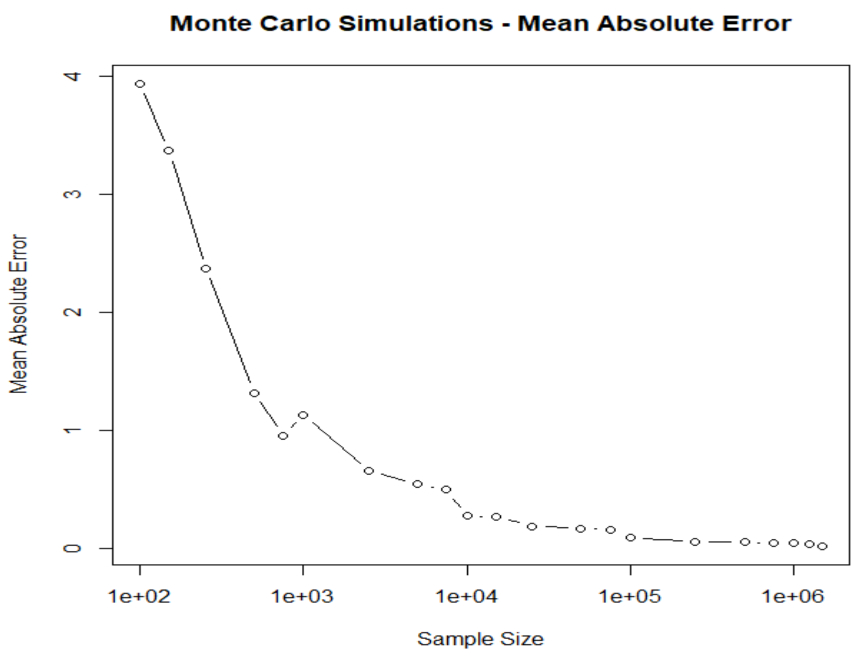
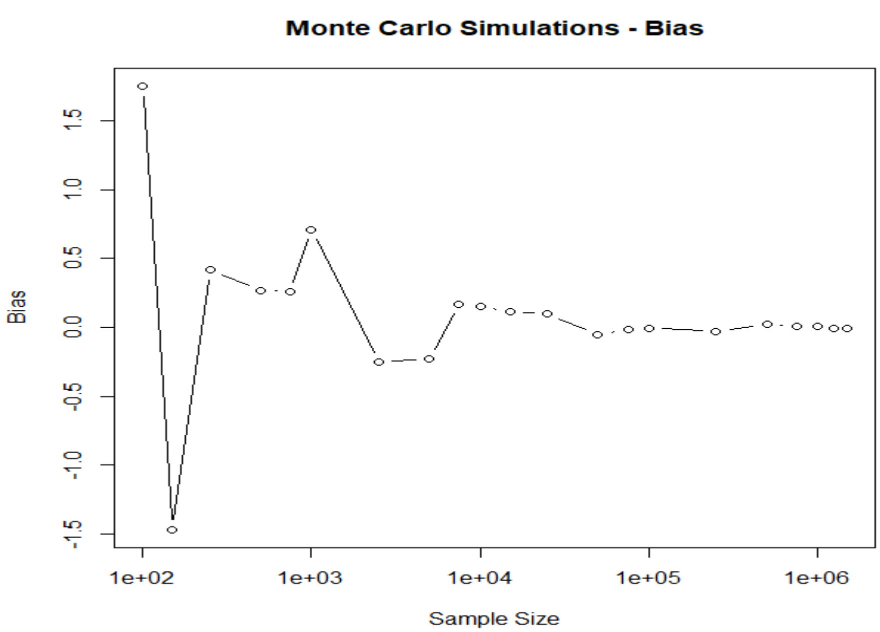


Figure 2



Conclusion

We explored veridical-type paradoxes using conditional probabilities, Bayes' rule, and mainly with Monte Carlo simulations. We discovered why a lot of discussion of the Monty Hall problem took place in 90's, and why so many researchers thought that the probability would be 50 per cent [7]. Those 50 per cent represent a 'flip a coin' decision. We discovered that contestants should change their minds, and it is the best decision among the explored options.

If contestants do not change their minds, it is the worst decision ever. Second worst decision is, if contestants are speculating too much – 'opposite tic-toc' decision. On the other hand, appropriate speculating is positive – 'tic-toc' decision, being the second-best decision. Research showed a lot about the Monty Hall problem and also about life. The aim of the study has been achieved.

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